Deepfake Image Detection Using Deep Learning Approaches

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***Abstract*—AI-manipulated facial image detection is increasingly challenging as new generators and manipulation methods continue to emerge. This study evaluates the generalization of six SOTA deepfake detection models from the ForensicHub framework on the Robust Deepfake Detection Challenge (RDDC) dataset. In addition to the ForensicHub-based experiments, a replication study of the winning RDDC 2026 solution, ShallowReal DINO-MAC, was conducted. To improve robustness, targeted dataset expansion, soft-voting ensembles, and a custom observation reweighting approach was proposed within the loss function. Effort model achieved the strongest baseline public test performance among the evaluated models, with 71% accuracy, 0.76 AUC, and 0.67 F1, while reweighting improved performance to 73% accuracy, 0.78 AUC, and 0.69 F1. Overall, the findings indicate that current detectors still fail to generalize under unseen conditions when training is constrained to a single consumer-grade GPU. However, targeted loss reweighting returns better results than baseline- and ensemble-based models. The source code is publicly available at https://github.com/ripsimes/Deepfake.**

***Index Terms*—deepfake detection, image classification, deep learning, image forensics, ensemble learning, loss reweighting**

I. Introduction

The rapid spread of synthetic and manipulated facial images raises identification risks in national security, misinformation, identity fraud, and cyberattacks. Existing state-of-the-art deepfake detection methods often show limited robustness to previously unseen generators, especially under unfamiliar manipulations or post-processing distortions. This work evaluates detector generalization on the Robust Deepfake Detection Challenge (RDDC) [1] dataset and explores targeted strategies to improve robustness.

The main contributions of this work are fourfold. Firstly, six representative deepfake detection models from the ForensicHub [2] framework are evaluated on the RDDC dataset. Second, the training set is expanded using additional benchmark data selected according to observed misclassification patterns. Third, soft-voting ensembles are evaluated by combining Effort [3] with other well-performing detectors. Fourth, TruFor-guided [4] loss reweighting is applied to the best-performing detector Effort. In addition, a small-scale replication of the winning DINO-MAC [5] solution is included as a comparative experiment. The architecture and parameters explicitly described in the report were reproduced, with reduced settings when required by GPU memory limitations.

II. Literature

Deepfake image detection has evolved from standard convolutional classifiers toward models designed to capture more robust forensic cues. Early backbone-based models [6] provide strong visual feature extraction baselines, while specialized detectors focus on face-forgery-specific artifacts [7], frequency-domain inconsistencies [8] , or reconstruction-based evidence [9]. More recent approaches emphasize generalization, since detectors trained on limited manipulation types often overfit to known fake patterns and perform poorly on new generation methods or degraded images. ForensicHub provides a unified benchmark and codebase for fake image detection and localization across multiple image domains, including deepfake detection, image manipulation detection and localization (IMDL) on natural images, AI-Generated image detection (AIGC), and document image manipulation localization. It integrates 42 baseline models, 6 backbones, and 23 datasets. In this study, ForensicHub was used as the baseline framework for selecting and evaluating six representative detectors Effort [3], TruFor [4], ResNet [6], Capsule-Net [7], SPSL [8], RECCE [9]. The RDDC, DF40 [10], and HiDF [11] datasets support evaluation across manipulation types and degradation conditions, making an assessment on real-world robustness. Details of the models and the progress of the field can be found in the ForensicHub [2] benchmark and the references therein.

III. Data

The experiments used the RDDC training, validation, and public test splits. The splits were equally balanced 50/50 between real and fake images. All images of the dataset are cropped facial images from CelebV-HQ [12] with some space around them. Only face-swapping and reenactment methods were applied. The training dataset contains 1000 samples, created with the FaceSwap method. Following the challenge recommendation, additional public data were incorporated to improve generalization. The degradation techniques included Gaussian noise, JPEG compression, smoothing, resizing, and color and contrast adjustments. The validation dataset contains 100 samples, created with the FaceSwap and StyleFeatureEditor methods. In addition to the set of degradation techniques used in the training set, speckle and Poisson noise were applied. The public test dataset contains 1000 samples,

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created with previous and FSGAN methods. Prior degradation techniques were used along with some less common salt-and-pepper noise, black-and-white color filters, and overlay addition. Fig. 1 shows real and fake samples from the RDDC public test set. In addition to the degradations already included in RDDC, the ForensicHub training pipeline applied probabilistic data augmentations during training, including horizontal and vertical flipping, brightness and contrast adjustment, image compression, 90-degree rotation, and Gaussian blur. These augmentations were applied only to the training data, while validation and testing used fixed resizing, preprocessing, and normalization to ensure consistent evaluation.

To support the targeted expansion of the training set, additional images from DF40 [10] and HiDF [11] were included in selected experiments. DF40 is a large-scale deepfake dataset containing diverse manipulation methods, and in this work it was used as the source of additional fake images from FaceSwap, DeepFaceLab, SimSwap, E4S, and InSwap, and real images from WhichFaceIsReal subsets. HiDF provides high-quality real and manipulated facial images designed to be difficult for human perception, and only its real-image subset was used here to increase authentic-face diversity.

IV. Methods

Six ForensicHub models were selected to cover both baseline and deepfake-specific detection strategies. One advantage of these models is that they can be deployed on a single consumer-grade GPU. Moreover, their implementations are publicly available. ResNet was included as a general backbone-based baseline, providing a simple reference point for comparison with more specialized detectors. Capsule-Net, SPSL, and RECCE were selected because they were originally designed for deepfake or face forgery detection and therefore represent established deepfake-oriented approaches within the framework. Effort was included because it was reported as the strongest generalizable AI-generated image detector, making it a relevant candidate for evaluating robustness on challenging public test data. TruFor was selected because its detection logic differs from standard classification-based models: it uses forensic trace analysis and anomaly localization to identify manipulation evidence, making it useful for comparison and for guiding the proposed loss-reweighting experiment.

A. Dataset Expansion

After analyzing misclassified samples from the initial detector outputs, recurring failure patterns were identified, including low-light faces, blur, compression artifacts, noise, overexposure, low-detail facial regions, and subtle manipulation traces. To address these cases, 1,220 additional real images were added to the original 500 real images from training split, using 600 real face images from WhichFaceIsReal, 300 real images from HiDF, 200 moderately degraded real images, and 120 harder real images generated from existing authentic samples through controlled blur, noise, brightness, compression, and low-detail resizing transformations. The fake class was expanded with 1,220 additional samples from DF40, including

Fig. 1. Real and fake samples from the RDDC public test set. Green borders indicate real images, and red borders indicate fake images.

340 FaceSwap, 270 DeepFaceLab, 225 SimSwap, 120 E4S, and 65 InSwap images, along with 200 degraded fake images generated using blur, noise, brightness shifts, overexposure, and JPEG compression. The original 1,000-image training set was expanded to a balanced 3,440-image training set containing 1,720 real and 1,720 fake images. This targeted expansion was designed to increase both artifact diversity and degradation robustness while preserving class balance in the final training set.

To explicitly address potential image degradation and artifact variations—such as blurring, compression, noise, and contrast shifts—a targeted degradation pipeline was applied to a subset of clean training samples. Rather than utilizing passively collected data, these images were actively subjected to controlled distortions, simulating severe post-processing conditions found in real-world scenarios. The composition, source distribution, and specific degradation parameters of this expanded training set are detailed in Table I, ensuring a balanced 50/50 split across authentic and manipulated categories.

TABLE I
Composition and Source Breakdown of the Expanded Training Dataset. Row colors distinguish image classes: denotes real images, and denotes fake images.

Source / Origin	Ct.	Description / Modification Parameters
Challenge Train	500	Original RDDC FaceSwap samples
WhichFaceIsReal	600	High-diversity real face images
HiDF Subset	300	High-quality authentic faces
Aug. Real (*augreal*)	200	Blur ($k \in \{3,5,7\}$), G. Noise ($\sigma \in [6,18]$), JPEG ($q \in [25,55]$), Darkening ($\alpha \in [0.55,0.9]$)
Hard Real (*hardreal*)	120	Blur ($k \in \{3,5,7\}$), G. Noise ($\sigma \in [6,22]$), JPEG ($q \in [20,55]$), Exposure ($\alpha \in [0.45,1.45]$), Scale ($\approx 33\%$)
Real Subtotal	**1,720**	**Balanced authentic pool**
Challenge Train	500	Original RDDC FaceSwap fakes
FaceSwap (DF40)	340	External face-swapping method
DeepFaceLab	270	Deep learning face reenactment
SimSwap	225	Identity-swapping framework
E4S	120	Style-based manipulation
InSwap	65	Alternative face-swapping pipeline
Degr. Fake (*degradedfake*)	200	Blur ($k \in \{3,5,7\}$), G. Noise ($\sigma \in [5,25]$), JPEG ($q \in [20,60]$), Exposure ($\alpha \in [0.3,2.0]$)
Fake Subtotal	**1,720**	**Balanced manipulated pool**
Full Train Set Total	**3,440**	**50/50 real-fake balance**

B. Soft-Voting Ensembles

To further improve detector performance, soft-voting ensembles were designed by pairing Effort with other trained detector and averaging their predicted probabilities. The averaged probability was thresholded at 0.5 to obtain the final label. TruFor, SPSL, and Capsule-Net were selected for pairing.

The ensemble probability for an input image x was computed as:

$$\hat{p}_{\mathrm{ens}}(x) = \frac{1}{M}\sum_{m=1}^{M} p_m(x) \tag{1}$$

where M is the number of models in the ensemble and $p_m(x)$ is the predicted fake probability from model m.

C. TruFor-Guided Effort Loss Reweighting

Effort is an image-level classifier that predicts whether an input image is real or fake, whereas TruFor provides richer localization-based forensic evidence. TruFor learns noise-sensitive forensic traces from real images in a self-supervised manner and identifies manipulations as deviations from authentic image patterns. As shown in Fig. 2, TruFor produces an anomaly map that highlights suspicious regions, a confidence map that estimates the reliability of the localization output, a Noiseprint++ representation that captures low-level forensic traces, and a final image-level integrity score derived from the anomaly and confidence information.

In this experiment, the pretrained TruFor model was kept frozen and used only to provide an image-level manipulation-evidence score for each training image. This score was used to reweight Effort's cross-entropy loss, increasing the contribution of images with stronger manipulation evidence during training. Since TruFor remained frozen, the reweighting acted as an external forensic guidance signal, while the trainable parameters belonged only to Effort.

The reweighted loss was defined as

$$\mathcal{L}_{\mathrm{reweighted}} = \frac{1}{N}\sum_{i=1}^{N} w_i \cdot \mathrm{CE}(z_i^{\mathrm{Effort}}, y_i), \tag{2}$$

where N is the batch size, z_i^{Effort} denotes Effort's logits for image i, y_i is the ground-truth label, and CE is the cross-entropy loss. For binary real/fake classification, the cross-entropy term is computed as

$$\mathrm{CE}(z_i^{\mathrm{Effort}}, y_i) = -\log \frac{e^{z_{i,y_i}^{\mathrm{Effort}}}}{\sum_{c=0}^{1} e^{z_{i,c}^{\mathrm{Effort}}}}, \tag{3}$$

where $z_{i,y_i}^{\mathrm{Effort}}$ is the logit corresponding to the ground-truth class.

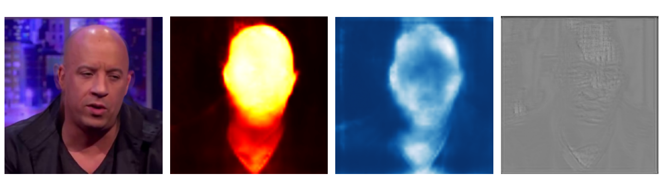

Fig. 2. TruFor localization outputs for a fake sample. The anomaly map highlights suspicious regions, the confidence map estimates their reliability, and Noiseprint++ represents low-level forensic traces.

The sample weight w_i was computed as

$$w_i = 1 + \alpha \cdot p_{\mathrm{TruFor}}(x_i), \tag{4}$$

where $p_{\mathrm{TruFor}}(x_i)$ is the manipulation-evidence score predicted by the frozen TruFor model for image x_i, and α controls the strength of the reweighting. The TruFor score is bounded as

$$p_{\mathrm{TruFor}}(x_i) \in [0, 1]. \tag{5}$$

In this study, α was set to 1, which constrained the sample weights to the range $[1, 2]$. This design ensured that every image continued to contribute to the optimization, while images with stronger TruFor manipulation evidence received higher importance in the loss calculation. For robustness analysis, additional experiments were conducted with $\alpha = 0.5$ and $\alpha = 2$.

D. DINO-MAC Replication Study

In addition to the ForensicHub-based experiments, a replication study of the winning RDDC 2026 solution, ShallowReal DINO-MAC, was conducted. The competition report describes the main architectural components, including a DINOv3-Large backbone with LoRA fine-tuning, a Multi-Aspect Classification (MAC) head, Dynamic Resolution training, Deep Supervision, Supervised Contrastive Learning (SCL), and Stochastic Depth.

The model was implemented incrementally by adding the reported components one at a time and retaining only configurations that improved public test performance. Due to GPU memory limitations, the backbone was scaled to DINOv3-Base, LoRA rank and alpha were reduced, and training resolutions were capped up to 384×384. Furthermore, intermediate deep supervision heads were omitted after ablation experiments showed they degraded validation performance. The experiments therefore represent an approximate replication of the published methodology rather than an exact reproduction of the winning submission, with the specific parameter differences detailed in Table II.

An additional experiment, DINO-MAC small + Effort Reweight, investigated whether the sample-weighting strategy learned by the proposed TruFor-guided Effort model could improve the replicated DINO-MAC detector. The DINO-MAC model was trained using per-image weights predicted by the pretrained Effort Adjusted Loss model. Thus, the weighting information transferred to DINO-MAC reflects the knowledge learned by the TruFor-guided Effort model rather than direct supervision from TruFor.

E. Metrics

Model performance was evaluated using Accuracy, Precision, Recall, and F1 score, defined in Eqs. (6)–(9), as well as AUC. Accuracy measures the overall proportion of correct predictions. Precision measures how many images predicted as fake are actually fake, while recall measures how many actual fake images are correctly detected. The F1 score summarizes the balance between precision and recall for the fake class.

TABLE II
COMPARISON OF ORIGINAL DINO-MAC SPECIFICATIONS AND REPLICATION PARAMETERS

Component / Parameter	**Original Report Specification**	**Replication Implementation**
Backbone Architecture	DINOv3-Large (1024-dim hidden state)	DINOv3-Base (768-dim hidden state)
LoRA Fine-Tuning	Rank $r = 32$, Alpha $\alpha = 64$	Rank $r = 16$, Alpha $\alpha = 32$
MAC Token Extraction	1 [CLS], 4 [REG], 1 [AVG]	1 [CLS], 4 [REG], 1 [AVG]
MLP Classifier Head	2 Fully-Connected layers, intermediate ReLU	2 Fully-Connected layers, intermediate ReLU, Dropout (0.2)
Training Resolution	Dynamic sampling $[384, 1152]$	Discrete batch sampling up to 384×384
Deep Supervision	Auxiliary heads on last 4 layers	Omitted (degraded validation performance during ablation)
Metric Learning	BCE + Supervised Contrastive Loss	BCE + Supervised Contrastive Loss

AUC evaluates the ranking quality of the predicted fake probabilities across different decision thresholds.

Let TP, TN, FP, and FN denote true positives, true negatives, false positives, and false negatives, respectively. In this work, the fake class is treated as the positive class. Accuracy, precision, recall, and F1 score were computed as:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}, \tag{6}$$

$$\text{Precision} = \frac{TP}{TP + FP}, \tag{7}$$

$$\text{Recall} = \frac{TP}{TP + FN}, \tag{8}$$

$$\text{F1} = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}. \tag{9}$$

These metrics were selected because the task requires both assigning correct binary labels and producing reliable confidence scores for distinguishing real and fake images.

All experiments, including model training, evaluation, and the dataset expansion pipeline, were executed utilizing a single NVIDIA T4 GPU with 16 GB of VRAM hosted on Google Colab. This hardware constraint ensures that the evaluated methodologies remain accessible and computationally feasible for deployment in resource-limited scenarios.

V. RESULTS AND DISCUSSION

The public test results are shown in Table III. Among the six baseline detectors, Effort achieved the strongest baseline performance, with 0.80 precision, 0.58 recall, 0.67 F1, 71% accuracy, and 0.76 AUC. The approximate DINO-MAC replication (Dino-Mac small) achieved 0.66 F1 and 0.73 AUC, showing competitive performance despite using reduced training settings. However, it did not outperform the best Effort-based reweighting result. These results confirm that detectors remain sensitive to unseen manipulation methods and degradation conditions.

The Effort model was further evaluated after targeted dataset expansion (Effort with Data Aug.). It achieved the second-highest recall of 0.62, with 70% accuracy, 0.75 AUC, and 0.67 F1. This suggests that dataset expansion improved exposure to additional artifacts and degradations, but did not fully solve public test generalization. The result indicates that simply adding more samples is not always sufficient; the added data must also match the difficulty of the target evaluation conditions.

The soft-voting ensemble results are also included in Table III. Among the tested combinations, Effort + Capsule-Net achieved the best ensemble performance, with 0.79 precision, 0.60 recall, 0.67 F1, 71% accuracy, and 0.76 AUC. This result matched the original Effort baseline in F1, accuracy, and AUC, but did not improve over the TruFor-guided loss reweighting approach. Effort + TruFor achieved the highest precision of 0.89 and an AUC of 0.76, but its low recall of 0.38 reduced its F1 score to 0.53. This means that the ensemble was cautious when predicting fake images, but it missed many actual fake samples. Effort + SPSL achieved 0.67 accuracy, 0.73 AUC, and 0.62 F1. These results suggest that ensemble voting is useful only when the paired detectors provide complementary and reliable probability estimates. If one model gives unreliable probability scores or makes systematic errors, averaging probabilities may reduce rather than improve performance.

The DINO-MAC replication result shows that the winning model design transfers reasonably well even under reduced hardware settings. The best replicated configuration, DINO-MAC small, achieved 0.66 precision, 0.66 recall, 0.66 F1, 66% accuracy, and 0.73 AUC. This model was not an exact reproduction of the winning solution, since the original configuration used DINOv3-Large and higher-resolution Dynamic Resolution training, while the reported experiment used a smaller backbone and lower resolutions due to GPU memory constraints. The DINO-MAC + Effort R. variant increased precision to 0.78 but reduced recall to 0.46, suggesting that Effort-based weighting made the DINO model more conservative rather than more balanced.

The best overall result was obtained using TruFor-guided

TABLE III
OVERALL PUBLIC TEST PERFORMANCE. ROW COLORS DISTINGUISH METHOD CATEGORIES: ■ DENOTES BASELINE DETECTORS, ■ DENOTES SOFT-VOTING ENSEMBLES, AND ■ DENOTES DINO-MAC REPLICATION EXPERIMENTS. BEST VALUES ARE BOLDED, AND SECOND-BEST VALUES ARE UNDERLINED.

Model / Method	**P**	**R**	**F1**	**Acc**	**AUC**	**Avg**
RECCE [9]	0.53	0.28	0.36	0.51	0.52	0.44
TruFor [4]	0.64	0.09	0.16	0.52	0.53	0.39
ResNet [6]	0.60	0.45	0.51	0.57	0.59	0.54
Capsule-Net [7]	0.53	**0.70**	0.61	0.54	0.57	0.59
SPSL [8]	0.65	0.49	0.55	0.61	0.65	0.59
Effort [3]	0.80	0.58	<u>0.67</u>	<u>0.71</u>	<u>0.76</u>	0.70
Effort with Data Aug.	0.74	0.62	<u>0.67</u>	0.70	0.75	0.70
Effort + TruFor	**0.89**	0.38	0.53	0.66	<u>0.76</u>	0.64
Effort + SPSL	0.74	0.54	0.62	0.67	0.73	0.66
Effort + Capsule-Net	0.79	0.60	<u>0.67</u>	<u>0.71</u>	<u>0.76</u>	<u>0.71</u>
DINO-MAC [5] small	0.66	<u>0.66</u>	0.66	0.66	0.73	0.67
DINO-MAC small + Effort R.	0.78	0.46	0.58	0.66	0.73	0.64
Effort Adjusted Loss (ours)	<u>0.83</u>	0.60	**0.69**	**0.73**	**0.78**	**0.73**

Effort loss reweighting (Effort Adjusted Loss), which improved public test performance to 0.83 precision, 0.60 recall, 0.69 F1, 73% accuracy, and 0.78 AUC. This result outperformed the baseline Effort model, the soft-voting ensembles, and the approximate DINO-MAC replication. The improvement suggests that TruFor's forensic evidence was more effective when used to guide Effort training directly than when used to reweight the DINO-MAC model.

Robustness analysis on the reweighting parameter revealed that both $\alpha = 0.5$ and $\alpha = 2$ yielded nearly identical, slightly improved performance (74% accuracy, 0.70 F1, and 0.78 AUC), demonstrating that the loss reweighting mechanism remains highly stable across different evidence scaling factors.

The prediction examples in Fig. 3 further illustrate the behavior of Effort on the public test set. The predicted fake probabilities show that the detector is confident on some examples but uncertain or incorrect on others, especially when visual artifacts are subtle or when real images contain degradation patterns similar to manipulated images. This supports the motivation for the targeted dataset expansion and TruFor-guided loss reweighting strategies, both of which were designed to address difficult failure cases rather than only improve performance on easy examples.

Overall, the experiments show that the strongest improvement was obtained by modifying the training objective rather than only increasing the training set or averaging model predictions. The results also show that public test performance remains limited, even for the strongest model.

VI. Conclusion

This work evaluated the generalization of six ForensicHub-based deepfake detection models on the RDDC dataset and investigated targeted dataset expansion, soft-voting ensembles, TruFor-guided Effort loss reweighting, and an approximate DINO-MAC small scale replication. Effort was the strongest baseline detector, while the proposed TruFor-guided loss reweighting achieved the best overall performance.

The dataset expansion experiment showed that adding targeted real and fake samples improved exposure to recurring failure patterns, such as blur, compression, noise, and subtle manipulation artifacts. While the expanded Effort with Data Augmentation increased recall from 0.58 to 0.62, it yielded only marginal improvements in accuracy, AUC, and F1. Effort with Capsule-Net provided the strongest overall ensemble performance, whereas Effort + TruFor achieved the highest precision; however, ensemble methods yielded inconsistent results, with some combinations performing worse than the baseline.

The best overall performance was achieved with TruFor Adjusted Loss, which improved public test performance to 0.83 precision, 0.60 recall, 0.69 F1, 73% accuracy, and 0.78 AUC. This suggests that localization-based forensic evidence from TruFor can provide useful external guidance for training an image-level classifier by increasing the loss contribution of samples with stronger manipulation evidence. The DINO-MAC small replication experiment achieved 0.66 F1 and 0.73 AUC, but did not exceed the Effort-based reweighting result. Since the exact training setup exceeded the available GPU memory, this experiment should be interpreted as an approximate replication of the reported method rather than a full reproduction.

Fig. 3. Effort prediction examples on the public test set. Border color indicates the ground-truth class: green for real and red for fake. The values below the images show the predicted fake probability.

All experiments were intentionally conducted on a single consumer-grade GPU to demonstrate the practicality and accessibility of the proposed approach under modest computational resources. Future work will investigate broader hyperparameter tuning, larger-scale training, and additional model combinations enabled by more extensive computational resources. In addition, the proposed reweighting strategy used only TruFor's image-level score, while its anomaly and confidence maps were not directly integrated into Effort's architecture. Future work should therefore explore the direct use of TruFor localization outputs as auxiliary supervision, including anomaly maps and confidence maps, rather than relying only on the final image-level score.

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